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Retinal Arteries and Veins Semantic Segmentation Using Modified U-Net Architecture

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***Abstract*—This paper presents a modified U-Net architecture designed for the semantic segmentation of retinal arteries and veins using infrared reflectance retinal imaging. Leveraging the RAVIR dataset, which comprises 46 high-quality retinal images, we focus on addressing the challenges of low contrast and varying lighting conditions often found in retinal images. Our approach incorporates advanced features such as bilinear interpolation and attention mechanisms to enhance segmentation accuracy. The modified U-Net architecture achieves a mean Intersection over Union (mIoU) of 0.8855 and a Dice coefficient of 0.9382, showing a significant improvement over the traditional U-Net. These results demonstrate its potential for practical applications in retinal disease diagnosis and monitoring.**

***Index Terms*—deep learning, U-Net, medical image analysis, infrared reflectance imaging, semantic segmentation, attention mechanisms**

I. Introduction

The segmentation of retinal veins and arteries is a critical task in ophthalmology, providing vital information for diagnosing and monitoring various retinal diseases. Accurate segmentation of these vascular structures enables quantitative analysis, crucial for assessing the severity and progression of conditions such as diabetic retinopathy, glaucoma, and age-related macular degeneration. Given the retina's delicate and complex structure, precise identification and differentiation of veins and arteries are essential for effective clinical assessment and treatment planning [1].

The retina, a thin layer of tissue at the back of the eye, plays a key role in vision. It contains a network of veins and arteries that supply blood to the retinal tissues, allowing proper retinal function (Fig. 1). Retinal vascular segmentation distinguishes between the two main types of blood vessels in the retina: veins, which return deoxygenated blood to the heart, and arteries, which supply oxygenated blood to the retinal tissues [2]. This differentiation is essential for diagnostic purposes, such as detecting retinal vein occlusions, arteriolar narrowing, and other vascular abnormalities indicative of systemic and ocular diseases, which, if left untreated, can lead to vision loss [3].

Traditional methods for retinal vessel segmentation relied heavily on manual annotation by clinicians or semi-automated techniques, which are time-consuming, subject to inter-observer variability, and often lack the accuracy needed for clinical use [4]. However, recent advancements in deep learning have revolutionized the field of medical image analysis, offering new avenues for automated and highly accurate segmentation [5], [6]. Convolutional Neural Networks (CNNs), a class of deep learning algorithms, have shown remarkable performance in various image segmentation tasks [7]. In the context of retinal vessel segmentation, CNNs can automatically extract relevant features from retinal images, enabling the precise delineation of veins and arteries without manual intervention [5]. The U-Net architecture has been particularly successful in medical image segmentation [8]. Its encoder-decoder structure, with skip connections, allows it to effectively capture both high-level and detailed features. However, while U-Net has shown promising results, there are still challenges, such as low contrast in certain areas of the retina, varying lighting conditions, and the presence of pathological features [9].

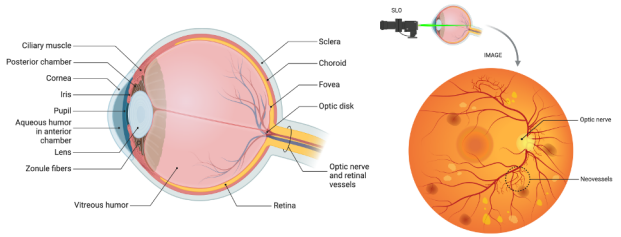

Fig. 1. Anatomy of human eye (left) and infrared reflectance imaging outline (right).

While the U-Net architecture has shown promising results, some challenges remain in handling low contrast areas, lighting variations, and differentiating between arteries and veins, particularly in the presence of pathologies. This paper presents key modifications to the standard U-Net architecture to specifically address these challenges. Our contributions include:

1) Incorporation of bilinear interpolation in upsampling layers to improve the accuracy of boundary reconstruction and reduce artifacts.
2) Integration of attention mechanisms, allowing the model to focus on critical features within the image and en-

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hance segmentation precision in challenging areas.

3) A customized loss function that combines cross-entropy loss, Dice loss, and a misclassification penalty, designed to mitigate class imbalance and reduce errors in artery-vein differentiation.

These innovations make the proposed architecture particularly well-suited for retinal image segmentation, requiring precise differentiation of fine vessel structures, even under challenging imaging conditions. We demonstrate that our modifications result in significant improvements in segmentation performance, achieving superior accuracy compared to the traditional U-Net and other state-of-the-art models. The structure of this paper is as follows: Section II reviews related work in retinal vessel segmentation. Section III describes our proposed methodology, including the modified U-Net architecture and training procedures. Section IV presents the experimental results and comparative analysis. Finally, Section V concludes the paper and discusses potential directions for future research.

II. Related Work

The segmentation and analysis of retinal vessels play a critical role in diagnosing and monitoring various ocular and systemic diseases. Recent advancements in deep learning and computer vision have significantly enhanced the capabilities of automated retinal image analysis. This literature review provides an overview of the key developments in this field, focusing on deep learning methods, semi-supervised approaches, and the challenges in segmentation and classification.

Recent advancements in deep learning have greatly improved the accuracy of retinal vessel segmentation. Deep learning architectures have demonstrated remarkable effectiveness in delineating vascular structures within retinal images, improving segmentation accuracy, robustness to noise, and adaptability to different imaging modalities [5]. These techniques have proven superior to traditional methods, particularly in handling complex vascular patterns and increasing the reliability of automated systems. A novel approach for simultaneous segmentation and classification of retinal arteries and veins, which decomposes these tasks into three related segmentation problems—arteries, veins, and vessels—was presented in [10]. This method showed superior performance, especially in managing complex vessel crossings. Semi-supervised learning approaches have emerged as effective solutions to the challenges posed by limited labeled data in medical imaging. In [11], the authors introduced a method combining labeled and unlabeled data to enhance model generalization and accuracy. Their semi-supervised approach leverages both supervised and unsupervised paradigms, achieving performance comparable to fully supervised models while requiring fewer annotated samples. This innovative strategy offers a cost-effective solution to the scarcity of annotated medical images, demonstrating its potential to improve segmentation outcomes in retinal imaging.

Understanding the clinical implications of retinal vascular diseases is essential for developing effective segmentation algorithms. A study provides insights into the structure and pathologies of retinal vessels, discussing conditions such as diabetic retinopathy and hypertensive retinopathy [3]. Their work emphasizes the role of accurate vessel segmentation in facilitating early diagnosis and treatment planning, highlighting the clinical significance of automated image analysis systems. By linking technical advancements to clinical applications, their research underscores the importance of integrating clinical knowledge into algorithm development. The evolution of retinal vessel segmentation methods is well-documented in the review, which compares traditional and deep learning techniques [12]. Classic model-based and statistical approaches often struggle with complex structures and noise, whereas deep learning methods excel in capturing intricate vessel patterns. This historical perspective offers valuable context for understanding the evolution of segmentation research and highlights the transformative impact of deep learning in overcoming the limitations of earlier methods. A novel approach that integrates CNNs with graph propagation techniques for simultaneous segmentation and classification of retinal vessels was presented in [13]. This method addresses both tasks in a unified framework, reducing computational complexity and improving both segmentation and classification accuracy. By distinguishing between arteries and veins, this joint approach enhances the clinical utility of automated retinal image analysis systems, paving the way for more comprehensive diagnostic tools.

Despite significant advancements, several challenges remain in retinal vessel segmentation. A review explores these challenges, highlighting the need for robust algorithms capable of handling diverse imaging conditions and integrating multimodal data [14]. They discuss the importance of developing environmentally sustainable and computationally efficient solutions, given the increasing demand for real-time processing in clinical settings. Their review identifies knowledge gaps and open issues, suggesting future research directions that emphasize the interpretation of complex vascular networks and improving algorithm scalability.

The segmentation of arteries and veins represents an extension of retinal vessel analysis, requiring models to classify vessel types accurately. Minimalistic models like W-Net have shown competitive performance in this task, matching the state-of-the-art with far fewer parameters [15]. The ability to segment and classify vessels efficiently has important implications for clinical applications, such as assessing artery-to-vein ratios and blood flow analysis. Furthermore, the application of minimalistic models to optical coherence tomography angiography (OCTA) images demonstrates their versatility across different imaging modalities. These models have been effective in segmenting complex vascular structures within OCTA images, providing a promising direction for future research in retinal imaging.

In conclusion, the reviewed literature demonstrates substantial progress in retinal vessel segmentation, driven by deep learning innovations and hybrid approaches. While considerable advancements have been made, ongoing research continues to address challenges such as data scarcity, computational efficiency, and clinical validation. By bridging the gap

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between technical innovation and clinical application, future developments in this field hold the potential to transform retinal disease diagnosis and management.

III. Materials and Methods

A. RAVIR Dataset

The RAVIR dataset is designed for the semantic segmentation and quantitative analysis of retinal arteries and veins [16]. It consists of 46 retinal images captured using infrared Scanning Laser Ophthalmoscopy (SLO), a form of infrared reflectance (IR) imaging. This imaging technology offers higher quality and contrast, is patient-friendly, and is less affected by optical media opacities and pupil size. The dataset includes images from both healthy and pathological cases, featuring subjects whose ages range from 19 to 88 years, with 29 males and 17 females. The images are annotated with arteries (label 128), veins (label 256), and background (label 0) [17].

*1) **Data Preprocessing**:* The dataset was divided into training and testing subsets. The dataset images are maintained at their original size of 768×768 pixels. Images were normalized to a range of [0, 1] by dividing pixel values by 255. This step helps in faster convergence during training by ensuring that the input data has a consistent scale.

*2) **Data Augmentation**:* To enhance the model's robustness and generalization capability, various data augmentation techniques were applied to the training images. These augmentations aim to simulate real-world variability and increase the diversity of the training set:

- **Horizontal and Vertical Flips:** Applied with a 50% probability to introduce variability in orientation.
- **Rotations:** Random rotations within a range of -30 to +30 degrees to account for rotational differences in retinal images.
- **Brightness and Contrast Adjustments:** Randomly applied to simulate different lighting conditions and improve model adaptability.
- **Random Gamma Correction:** Introduced with a 20% probability to adjust the gamma levels and enhance contrast perception.
- **Blurring:** Applied with a kernel size between 3 and 7 to introduce blur and mimic out-of-focus conditions.
- **Shift, Scale, and Rotate:** Random transformations including shifts (up to 10%), scaling (up to 10%), and additional rotations to further increase diversity.
- **Elastic Transformations:** Applied to simulate elastic deformations in the images, helping the model learn to cope with structural changes.
- **Grid Distortion:** Used to introduce non-linear distortions by varying the grid structure, enhancing the model's ability to handle geometric variations.

These augmentations were implemented with an 'augmentation_times' parameter set to 8, meaning each image was augmented eight times during training to provide a wide range of variations for the model to learn from.

*3) **Dataset Splitting**:* The RAVIR dataset used in our experiments initially consists of 23 original retinal images. These images are divided into training and testing subsets using an 80:20 split, resulting in 18 images allocated for training and 5 images reserved for testing. This split ensures that the model's performance is assessed on unseen data, providing a reliable evaluation metric.

To enhance the training process, we apply data augmentation techniques to the 18 training images. The augmentations are performed dynamically during training, effectively expanding the training dataset to 144 images. This process ensures that the model is exposed to a diverse range of variations for each training image, thus improving its generalization capabilities. The testing set of 5 images remains unaugmented, serving as a consistent baseline to evaluate model performance.

B. Modified U-Net Architecture

For the segmentation task, we utilized a customized U-Net architecture tailored to meet the specific demands of retinal image segmentation. Our model integrates several enhancements to the traditional U-Net framework [8], known for its efficacy in biomedical image segmentation. Its unique 'U' shape, formed by a contracting path and an expansive path, is specifically structured to capture both context and details. This customized architecture designed to improve feature extraction, segmentation accuracy, and overall model performance. The modifications focused on incorporating several key enhancements:

- **Bilinear Interpolation:** We incorporated bilinear interpolation in the upsampling layers to enhance boundary reconstruction, ensuring smoother transitions and more accurate delineation of retinal structures.
- **Attention Mechanisms:** We integrated attention mechanisms into the U-Net architecture to improve the model's ability to focus on important features and suppress irrelevant background noise. The attention modules help the network selectively enhance critical features, leading to improved segmentation accuracy and consistency in detecting arteries and veins.
- **Encoder and Decoder Improvements:** We made targeted improvements to the encoder and decoder paths, including adjustments to convolutional layers, activation functions, and skip connections. These enhancements ensure more effective feature extraction and reconstruction, improving overall segmentation performance.

These modifications collectively enhance the model's ability to accurately segment retinal images, effectively addressing the challenges posed by low contrast, varying lighting conditions, and the presence of pathological features.

C. Bilinear Interpolation

Bilinear interpolation is used in the upsampling layers of the U-Net model for the segmentation of retinal veins and arteries. This method offers several advantages that enhance the quality of segmentation, model performance, and computational efficiency.

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1) Improved Segmentation Quality:

- **Smoothness:** Bilinear interpolation can provide smoother and more continuous predictions, particularly important for the segmentation of continuous structures like veins and arteries.
- **Reduced Artifacts:** It minimizes the risk of introducing checkerboard artifacts, which can be problematic in the precise delineation of retinal vessels.

2) Enhanced Model Performance:

- **Consistency:** Bilinear interpolation ensures a consistent upscaling that is especially beneficial for the fine and intertwined structures of retinal vessels.
- **Accuracy:** By preserving the continuity of vessel structures, bilinear interpolation can potentially increase the accuracy of segmentation for both thick and thin vessels.

3) Computational Efficiency: Unlike transpose convolution, bilinear interpolation does not introduce additional learnable parameters, which reduces the model complexity and thus, the computational load.

4) Ease of Training: **Stable Gradients:** Bilinear interpolation provides stable gradients during training, which can make the learning process smoother and possibly require less tuning of hyperparameters.

D. Attention Mechanisms

Attention mechanisms [18] were integrated into the U-Net model to enhance the network's ability to focus on relevant features while minimizing the influence of background noise. In retinal images, where low-contrast regions and lighting variability can make the fine differentiation between arteries, veins, and background structures challenging. Attention mechanisms allow the model to selectively amplify significant regions of the feature maps and suppress less relevant or noisy information, leading to more accurate segmentation [19].

In our architecture, **Channel Attention** is applied after each upsampling step in the decoder path, following the concatenation of feature maps from the encoder and decoder. This enables the model to prioritize the most informative features for segmentation, particularly in regions where arteries and veins are difficult to distinguish.

*1) **Channel Attention Mechanism**:* We specifically adopted **channel attention** [20] to highlight the most informative feature channels at each stage of the upsampling process. Each channel in a feature map represents different aspects of the image, such as edges, textures, or regions of interest [19]. The channel attention mechanism, as shown in Fig. 3, assigns different weights to each feature channel, which helps the model focus more on the channels that are critical for distinguishing between retinal arteries and veins. This mechanism works by:

- **Global Average Pooling and Global Max Pooling**: For each feature map, the model computes two global context representations using both average pooling and max pooling. These operations summarize the importance of each channel across the entire spatial dimension.
- **Fully Connected Layers**: The pooled feature maps are passed through a small neural network (two convolutional layers) that generates channel-wise attention weights. These weights represent the importance of each channel in the current feature map.
- **Weighted Feature Maps**: The attention weights are multiplied with the original feature maps, amplifying the most important channels and suppressing less informative ones. This allows the model to better focus on relevant vascular structures.

The attention mechanisms improve segmentation by focusing on vascular structures, reducing false positives, and handling varying lighting conditions. By assigning more weight to channels that represent important features, such as arteries and veins, the attention mechanism allows the model to better differentiate between vessels and background noise. This leads to fewer false positives, particularly in regions where arteries and veins cross or overlap, by emphasizing only the most critical channels. Additionally, the attention mechanism helps the model dynamically adjust its focus in areas with varying lighting or low contrast, ensuring accurate vessel segmentation. By applying channel attention at key stages of the U-Net model, we are able to selectively enhance important features, significantly improving segmentation performance in retinal images where arteries and veins often present complex and variable patterns.

E. Loss Function

To optimize the performance of our modified U-Net architecture for retinal vessel segmentation, we designed an enhanced loss function. This function combines Cross-Entropy Loss, Dice Loss, and a specific penalty term to address both general accuracy and targeted misclassification issues.

- **Cross-Entropy Loss:** This component of the loss function focuses on pixel-wise classification accuracy by measuring the discrepancy between the predicted class probabilities and the true class labels. It is effective in guiding the model to learn precise pixel classifications. The Cross-Entropy Loss is formulated as:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(p_{i,c}) \tag{1}$$

  where N is the number of pixels, C is the number of classes, $y_{i,c}$ is the true binary indicator for class c at pixel i, and $p_{i,c}$ is the predicted probability for class c at pixel i.

- **Dice Loss:** The Dice Loss is incorporated to optimize the overlap between the predicted segmentation and the ground truth. It is particularly beneficial for handling class imbalance, ensuring that smaller and less frequent structures are accurately segmented. The Dice Loss is expressed as:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i + \epsilon} \tag{2}$$

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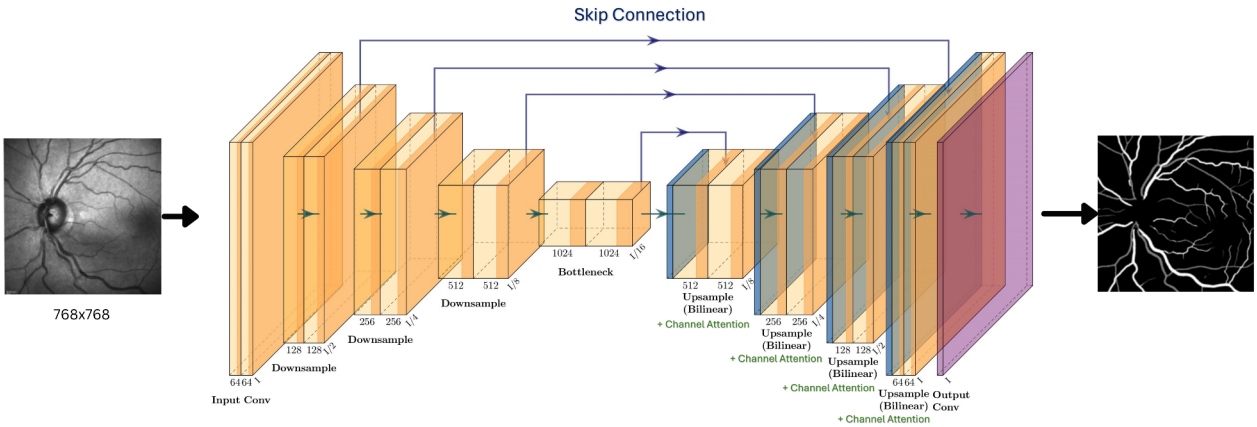

Fig. 2. The modified U-Net architecture included the use of bilinear interpolation in the upsampling layers and Channel Attention, for more precise boundary reconstruction.

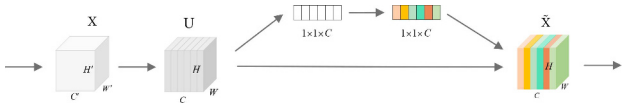

Fig. 3. A visual representation of channel attention [21]

where y_i and p_i are the ground truth and predicted values for pixel i, respectively, and ϵ is a small constant for numerical stability.

- **Specific Misclassification Penalty:** To further refine the segmentation quality, a penalty term is introduced for specific misclassifications. This term adds a penalty when pixels of class 1 are incorrectly classified as class 2, which is critical in retinal vessel segmentation where such misclassifications can lead to significant errors. The penalty is defined as:

$$\mathcal{L}_{Penalty} = \lambda \left(\frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[(\hat{y}_i = 1) \wedge (y_i = 2)] \right) \quad (3)$$

where $\mathbb{I}[\cdot]$ is the indicator function that evaluates to 1 if the condition is true and 0 otherwise, and λ is the penalty weight.

The loss function used for training is a weighted sum of these components:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{CE} + \beta \mathcal{L}_{Dice} + \gamma \mathcal{L}_{Penalty} \quad (4)$$

In our experiments, the weights α, β, and γ were set to 0.5, 0.25, and 0.25, respectively, balancing the contributions of each component to optimize overall model performance.

F. Optimizer and Learning Scheduler

To optimize the U-Net model for segmenting retinal veins and arteries, we employ the Adam optimizer. Adam [22] is chosen for its adaptive learning rate capabilities, which combine the advantages of two other popular optimization algorithms: AdaGrad and RMSProp. This helps in efficient and effective training by adjusting the learning rates of individual parameters.

In addition to the Adam optimizer, we use a learning rate scheduler to dynamically adjust the learning rate during training. This helps in fine-tuning the training process, allowing for larger updates when the model is far from the optimal solution and smaller updates as it converges. The Cosine Annealing Scheduler adjusts the learning rate following a cosine function. This approach smoothly decreases the learning rate from an initial value to a predefined minimum over a set period (T_{max}), which encourages exploration early in training and fine-tuning towards convergence. The scheduler is defined by the equation:

$$\eta(t) = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}}) \left(1 + \cos\left(\frac{t\pi}{T_{\text{max}}} \right) \right), \quad (5)$$

where $\eta(t)$ is the learning rate at epoch t, η_{max} is the initial learning rate, η_{min} is the minimum learning rate, and T_{max} is the period over which the learning rate anneals.

In our experiments, the Cosine Annealing Scheduler was configured with $T_{\text{max}} = 200$ epochs and $\eta_{\text{min}} = 0.000001$. This setup provides a smooth transition in learning rates throughout the training process, enhancing the model's ability to generalize and effectively segment retinal vessels.

G. Training Process

In implementation, we initialize the Adam optimizer with a learning rate of 0.001 and use the Cosine Annealing scheduler (200 epochs, Minimum Learning Rate (η_{min}): 0.000001) to dynamically adjust the learning rate over the training process. This setup is designed to facilitate efficient training by allowing the learning rate to decrease smoothly, thus promoting effective convergence and preventing overfitting.

Using the Adam optimizer in conjunction with the Cosine Annealing learning scheduler allows the U-Net model to

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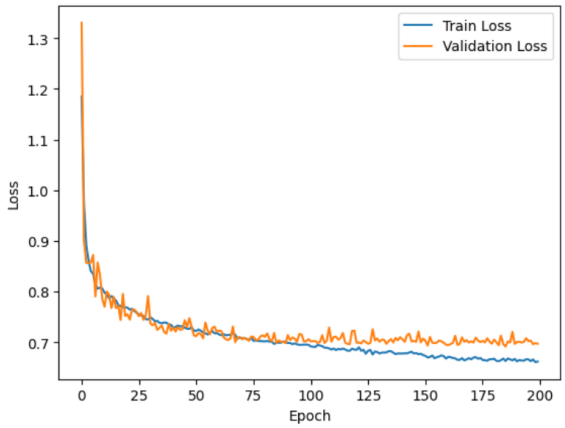

Fig. 4. Training and Validation Loss over 200 epochs, illustrating the impact of the Cosine Annealing scheduler on convergence.

effectively learn to segment retinal veins and arteries, as illustrated in Fig. 4. The adaptive learning rate provided by the scheduler enables smooth transitions between exploration and convergence, enhancing model performance and reducing the risk of overfitting.

H. Model Evaluation

To evaluate the performance of the segmentation model, we use the Dice coefficient and Mean Intersection over Union (mIoU).

The Dice coefficient measures the overlap between the predicted segmentation and the ground truth and is defined as:

$$\text{Dice coefficient} = \frac{2 \cdot |A \cap B|}{|A| + |B|}, \tag{6}$$

where A is the predicted segmentation and B is the ground truth segmentation.

The IoU is used to evaluate the segmentation performance for each class. It measures the overlap between the predicted and ground truth masks and is defined as:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}, \tag{7}$$

where A is the predicted segmentation and B is the ground truth segmentation. The mIoU is the mean IoU across all classes.

IV. Results and Discussions

The U-Net model was evaluated on both the training (Table I) and test datasets (Table II). One of the main challenges was working with a small dataset, as the RAVIR dataset contains only 46 retinal images, making it difficult to achieve good generalization. Despite the limitations in data availability, our model managed to perform well through the use of data augmentation and attention mechanisms. The results indicate the following performance:

The evaluation metrics and visual results demonstrate that the model has learned to segment retinal veins and arteries with high accuracy. The quantitative metrics, including mIoU and Dice coefficient, indicate strong performance on both the training and test sets. The visual results further confirm the model's ability to accurately delineate the vascular structures in retinal images, as shown in Fig. 5. We also deployed our model on the testing dataset, which is unlabeled, and obtained promising results as shown in Fig. 6 and Table II.

TABLE I
Evaluation Metrics for Retinal Veins and Arteries Segmentation on the Training Dataset.

Metric	Training Set	Validation Set
mIoU	0.795	0.728
Dice Coefficient	0.882	0.831

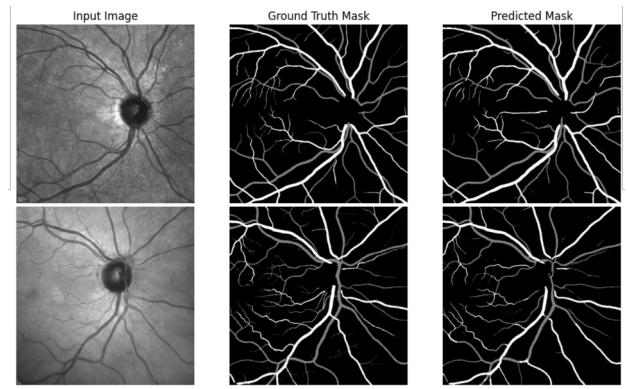

Fig. 5. Sample prediction on the validation dataset: Input image (left), ground truth mask (middle), and predicted mask (right).

TABLE II
Evaluation On the Testing Dataset after Training on 144 Images and Validating on 5 Images.

Metric	Value
Mean Dice	0.7631 ± 0.0479
Artery Dice	0.7545 ± 0.0486
Vein Dice	0.7716 ± 0.0496
Mean Jaccard	0.6194 ± 0.0629
Artery Jaccard	0.6081 ± 0.0624
Vein Jaccard	0.6307 ± 0.0669

A. Training without splitting the labeled data

Table III presents the evaluation metrics applied to the entire dataset, comparing our approach with other methods using the mean Dice coefficient and mIoU. After deployed our model on the testing dataset, we obtained promising results as shown in Fig. 7 and Table IV. These visual results provide a qualitative assessment of the model's ability to segment retinal veins and arteries, further validating the quantitative metrics presented above. The model's performance on the test dataset, despite being unlabeled, indicates its robustness and potential for real-world applications in medical image segmentation.

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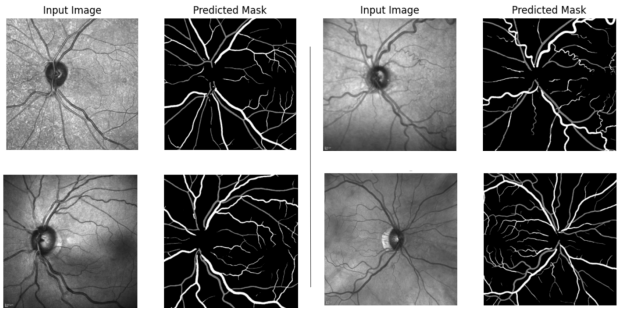

Fig. 6. Sample predictions on the test dataset: Input images (left) and corresponding predicted masks (right).

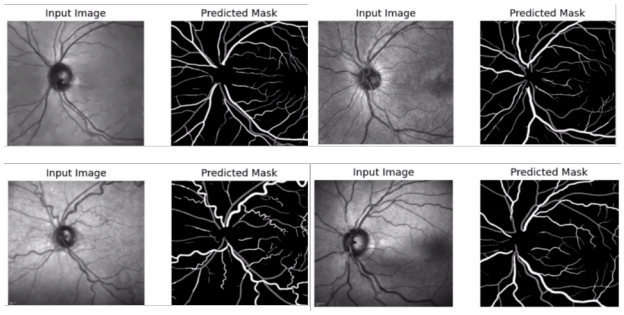

Fig. 7. Sample predictions on the test dataset: Input images (left) and corresponding predicted masks (right).

TABLE III
EVALUATION ON THE WHOLE LABELED DATA AND COMPARING WITH OTHER METHODS

Method	mIoU	Dice Coefficient
U-Net [23], [24]	0.6119	0.7592
Dense U-Net [24], [25]	0.6186	0.7644
Residual U-Net [24], [26]	0.6240	0.7685
R2U-Net [24], [26]	0.6291	0.7723
Recurrent U-Net [24], [26]	0.6385	0.7794
NexToU [27]	0.6422	0.7821
U-Net++ [24], [28]	0.6454	0.7845
DU-Net [24], [29]	0.6532	0.7902
AG-Net [24], [30]	0.6554	0.7918
IterNet [24], [31]	0.6602	0.7953
CE-Net [24], [32]	0.6653	0.7990
SegRAVIR [24]	0.7085	0.8294
cGAN [33]	0.7800	0.8764
Modified-UNet (OURs)	**0.8855**	**0.9382**

TABLE IV
EVALUATION ON THE TESTING DATASET AFTER TRAINING ON THE WHOLE DATASET, TRANSPOSED

Metric	Value
Mean Dice	0.7689 ± 0.0487
Artery Dice	0.7621 ± 0.0483
Vein Dice	0.7756 ± 0.0504
Mean Jaccard	0.6271 ± 0.0647
Artery Jaccard	0.6180 ± 0.0630
Vein Jaccard	0.6362 ± 0.0682

V. CONCLUSION

In this study, we have successfully demonstrated the efficacy of a modified U-Net architecture for the segmentation of retinal arteries and veins. Despite the challenge of working with a small dataset, our model achieved strong performance, with a mean Dice coefficient of 0.9382 and an mIoU of 0.8855, surpassing many state-of-the-art methods. By incorporating enhancements such as bilinear interpolation and attention mechanisms, our model effectively addresses the challenges posed by the complex and variable nature of retinal images. The experimental results showcase substantial improvements in segmentation accuracy, as evidenced by high mIoU and Dice coefficient values. These findings underscore the potential of our approach in advancing the automated analysis of retinal images, thereby facilitating more accurate and timely diagnosis of retinal diseases. Future work will focus on overcoming small dataset limitations through semi-supervised or transfer learning, exploring multimodal data integration for improved robustness, and optimizing the model for real-time clinical deployment by reducing computational complexity.

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